

CHEMISTRY — CHAPTER 10

ACIDS, BASES AND SALTS — QUICK PREP

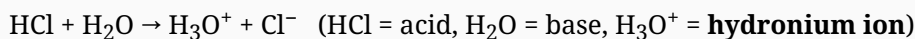
Paper 23 ke 50 marks mein se **35 marks sirf isi chapter se hain**. Baqi 15 marks revision ke hain (Ch 4, 6, 8, 9).

1. Do definitions jo poora chapter chalati hain

Arrhenius concept: Acid woh substance hai jo paani mein ionize ho kar **H⁺ ions** deta hai. Base woh hai jo paani mein ionize ho kar **OH⁻ ions** deta hai.



Bronsted-Lowry concept: Acid = **proton donor** (H⁺ deta hai). Base = **proton acceptor** (H⁺ leta hai).



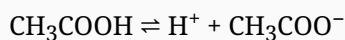
Arrhenius theory ki limitation: yeh sirf aqueous solutions par lagti hai. Yeh explain nahin kar sakti ke CO₂ aur SO₂ acidic kyun hain (in mein H hi nahin) aur NH₃ base kyun hai (is mein OH hi nahin). Isi liye 1923 mein Bronsted-Lowry theory aayi.

Amphoteric: jo substance acid aur base dono ki tarah react kare. **Paani amphoteric hai** — kabhi proton deta hai, kabhi leta hai.

2. Strong vs Weak (yeh sirf ionization ke baare mein hai)

Strong acid: paani mein **100% ionize** ho jaye. Examples: **HCl, HNO₃, H₂SO₄**. Single arrow (→) use karo.

Weak acid: poori tarah ionize na ho. Example: **ethanoic acid (CH₃COOH)** jo sirkay (vinegar) mein hota hai, sirf 5% tak ionize hota hai. Reversible arrow (⇌) use karo.



Strong bases: NaOH, KOH, Ca(OH)₂. **Weak bases:** Al(OH)₃, NH₃.

3. Base aur Alkali ka farq (yeh zaroor aata hai)

Bases = metals ke **oxides aur hydroxides**. **Alkali** = woh base jo **paani mein ghul jaye** (water soluble). Yani: **har alkali ek base hai, lekin har base alkali nahin hai**.

Alkali hain: NaOH, KOH, Ca(OH)₂. Alkali **nahin** hain: Cu(OH)₂, Al(OH)₃, Fe(OH)₃ (yeh paani mein nahin ghulte).

Ghar ki cheezein jin mein alkali hota hai: soap, detergent, shampoo, toothpaste.

4. Acids ki 5 properties (equations yaad karo)

1. Taste khatta (sour). 2. **Blue litmus ko red** kar dete hain.

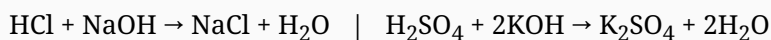
3. **Metal ke saath:** Metal + Acid → Salt + **Hydrogen gas**



4. **Metal carbonate ke saath:** Metal carbonate + Acid → Salt + Water + **Carbon dioxide**



5. **Base ke saath (neutralization):** Acid + Base → Salt + Water



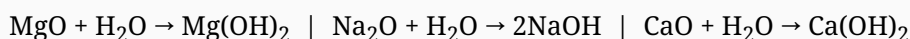
5. Bases ki 5 properties

1. Taste karwa (bitter).
2. **Red litmus ko blue** kar dete hain.
3. Solution chhoone par **slippery** lagta hai.
4. Acids ko **neutralize** karte hain → salt + water.
$$\text{H}_2\text{SO}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$$
5. Garam karne par **ammonium salts ko decompose** karte hain aur **ammonia gas** nikalti hai.
$$\text{NaOH} + \text{NH}_4\text{Cl} \rightarrow \text{NaCl} + \text{NH}_3 + \text{H}_2\text{O} \quad | \quad \text{KOH} + \text{NH}_4\text{Cl} \rightarrow \text{KCl} + \text{NH}_3 + \text{H}_2\text{O}$$

6. Oxides aur Hydroxides

Metal + Oxygen → Metal oxide $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ | $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$

Metal oxide + Water → Metal hydroxide



Amphoteric oxides/hydroxides: jo acid aur base dono ke saath react karte hain — Al_2O_3 , $\text{Al}(\text{OH})_3$, ZnO , $\text{Zn}(\text{OH})_2$.

7. Acid Rain

Normal baarish pehle se halki si acidic hoti hai kyunki us mein hawa ki CO_2 ghul jaati hai. Insani activity se bohat saray **acid oxides** hawa mein chale jaate hain, jo baarish ko **bohat zyada acidic** bana dete hain — isay **acid rain** kehte hain.

Acid rain mein ghula hua hota hai: **nitric acid (HNO_3)** aur **sulphuric acid (H_2SO_4)**.

Nuqsan: acids corrosive hote hain, is liye acid rain **buildings, structures aur statues** ko kha jaati hai — khaas taur par woh jin mein **metals** aur **metal carbonates** (marble, limestone) hote hain.

8. Traps — yeh pairs ulat mat karna

Yeh	Us se farq
Acid + metal → H_2 gas	Acid + carbonate → CO_2 gas
Strong acid = poora ionize hota hai	Concentrated ya "ghul jaata hai" — yeh strength nahin hai
Normal rain halki acidic = CO_2 ki wajah se	Acid rain = SO_2 aur NO_2 ki wajah se
Har alkali ek base hai	Har base alkali nahin ($\text{Cu}(\text{OH})_2$ paani mein nahin ghulta)
Acid solution mein H^+ zyada	Alkali solution mein OH^- zyada
Strong acid → single arrow	Weak acid ⇌ reversible arrow

9. Paper ke liye 3 process reminders

- Jahan **state symbols** maange hain wahan **(s), (l), (g), (aq)** zaroor likhna — exam-22 mein yeh mark gaya tha.
- Har equation **balance** kar ke check karna. Kitab mein kuch equations un-balanced chhapi hain, is liye khud ginna.
- Har calculation aur explanation ke aakhir mein **ek poora jumla** likhna jo sawal ka jawab de.

Video (NBF Federal Board, poora Chapter 10 exercise):

<https://www.youtube.com/watch?v=zELXCuDYFWI>